

SQL Practice Workbook (DDL → DML → DQL)

Scenario: Employee Management System

PART 1: DDL (Data Definition Language) – 20 Tasks

Task 1

Create a database named company_db.

Commands:

```
CREATE DATABASE company_db;
```

Task 2

Display all available databases.

Commands:

```
SHOW DATABASES;
```

Task 3

Select the company_db database.

Commands:

```
USE company_db;
```

Task 4

Create a table named employees with columns:

- emp_id
- employee_name
- department
- salary

Commands:

```
CREATE TABLE employees (  
    emp_id INT,  
    employee_name VARCHAR(50),  
    department VARCHAR(50),  
    salary DECIMAL(10,2)  
);
```

Task 5

Display the structure of the employees table.

Commands:

```
DESC employees;
```

Task 6

Add a new column email.

Commands:

```
ALTER TABLE employees ADD email VARCHAR(100);
```

Task 7

Add a new column joining_date.

Commands:

```
ALTER TABLE employees ADD joining_date DATE;
```

Task 8

Display the updated table structure.

Commands:

```
DESC employees;
```

Task 9

Modify the size of the employee_name column.

Commands:

```
ALTER TABLE employees MODIFY employee_name VARCHAR(100);
```

Task 10

Rename column employee_name to emp_name.

Commands:

```
ALTER TABLE employees RENAME COLUMN employee_name TO emp_name;
```

Task 11

Add a PRIMARY KEY on emp_id.

Commands:

```
ALTER TABLE employees ADD PRIMARY KEY(emp_id);
```

Task 12

Add a UNIQUE constraint on email.

Commands:

```
ALTER TABLE employees ADD CONSTRAINT unique_email UNIQUE(email);
```

Task 13

Add a CHECK constraint so salary must be greater than 10000.

Commands:

```
ALTER TABLE employees ADD CONSTRAINT chk_salary CHECK(salary > 10000);
```

Task 14

Rename table employees to employee_details.

Commands:

```
ALTER TABLE employees RENAME TO employee_details;
```

Task 15&16

Create another table named departments.

Add columns:

- dept_id
- dept_name

Commands:

```
CREATE TABLE departments( dept_id INT, dept_name VARCHAR(50) );
```

Task 17

Display all tables in the database.

Commands:

```
SHOW TABLES;
```

Task 18

Drop the departments table.

Commands:

```
DROP TABLE departments;
```

Task 19

Display all tables again.

Commands:

SHOW TABLES;

Task 20

Drop the employee_details table.

Commands:

DROP TABLE employee_details;

PART 2: DML (Data Manipulation Language) – 20 Tasks**Create Employee Table Again**

Before performing DML operations, recreate the table.

Task 21

Insert one employee record.

Commands:

INSERT INTO employees

VALUES (101,'Saran','IT',50000,'saran@gmail.com','2023-01-10');

Task 22

Insert another employee record.

Commands:

INSERT INTO employees

VALUES (102,'Priya','HR',45000,'priya@gmail.com','2023-02-15');

Task 23

Insert 3 employee records using a single query.

Commands:

INSERT INTO employees VALUES

(103,'Rahul','Finance',55000,'rahul@gmail.com','2023-03-01'),
(104,'Sneha','Marketing',48000,'sneha@gmail.com','2023-04-01'),
(105,'Kiran','Sales',52000,'kiran@gmail.com','2023-05-01');

Task 24

Insert employees belonging to IT department.

Commands:

```
INSERT INTO employees
```

```
VALUES (106,'Arun','IT',60000,'arun@gmail.com','2023-06-01');
```

Task 25

Insert employees belonging to HR department.

Commands:

```
INSERT INTO employees
```

```
VALUES (107,'Meena','HR',47000,'meena@gmail.com','2023-07-01');
```

Task 26

Insert employees belonging to Finance department.

Commands:

```
INSERT INTO employees
```

```
VALUES (108,'Vijay','Finance',58000,'vijay@gmail.com','2023-08-01');
```

Task 27

Insert employees belonging to Marketing department.

Commands:

```
INSERT INTO employees VALUES
```

```
(109,'Divya','Marketing',49000,'divya@gmail.com','2023-09-01');
```

Task 28

Insert employees belonging to Sales department.

Commands:

```
INSERT INTO employees
```

```
VALUES (110,'Ravi','Sales',51000,'ravi@gmail.com','2023-10-01');
```

Task 29

Update salary of employee with emp_id = 101.

Commands:

```
UPDATE employees SET salary = 55000 WHERE emp_id = 101;
```

Task 30

Update department of employee with emp_id = 105.

Commands:

```
UPDATE employees SET department = 'IT' WHERE emp_id = 105;
```

Task 31

Update email of employee with emp_id = 103.

Commands:

```
UPDATE employees SET email = 'rahul_new@gmail.com' WHERE emp_id = 103;
```

Task 32

Increase salary of all IT employees by ₹5000.

Commands:

```
UPDATE employees SET salary = salary + 5000 WHERE department = 'IT';
```

Task 33

Increase salary of all HR employees by 10%.

Commands:

```
UPDATE employees SET salary = salary * 1.10 WHERE department = 'HR';
```

Task 34

Change department from Sales to Marketing.

Commands:

```
UPDATE employees SET department = 'Marketing' WHERE department = 'Sales';
```

Task 35

Delete employee with emp_id = 110.

Commands:

```
DELETE FROM employees WHERE emp_id = 110;
```

Task 36

Delete all employees from Finance department.

Commands:

```
DELETE FROM employees WHERE department = 'Finance';
```

Task 37

Insert new Finance employees.

Commands:

```
INSERT INTO employees VALUES  
(111,'Karthik','Finance',62000,'karthik@gmail.com','2024-01-01'),  
(112,'Anitha','Finance',57000,'anitha@gmail.com','2024-01-05');
```

Task 38

Delete employees whose salary is less than ₹45000.

Commands:

```
DELETE FROM employees WHERE salary < 45000;
```

Task 39

Update all Marketing employee salaries by ₹3000.

Commands:

```
UPDATE employees SET salary = salary + 3000 WHERE department = 'Marketing';
```

Task 40

Delete all records from the table using DELETE.

Commands:

```
DELETE FROM employees;
```

PART 3: DQL (Data Query Language) – 20 Tasks

After completing DML tasks, insert fresh employee data and perform the following queries.

Task 41

Display all employee records.

Commands:

```
SELECT * FROM employees;
```

Task 42

Display only employee names.

Commands:

```
SELECT emp_name FROM employees;
```

Task 43

Display employee names and salaries.

Commands:

```
SELECT emp_name, salary FROM employees;
```

Task 44

Display employees working in IT department.

Commands:

```
SELECT * FROM employees WHERE department='IT';
```

Task 45

Display employees working in HR department.

Commands:

```
SELECT * FROM employees WHERE department='HR';
```

Task 46

Display employees with salary greater than ₹50000.

Commands:

```
SELECT * FROM employees WHERE salary > 50000;
```

Task 47

Display employees with salary less than ₹60000.

Commands:

```
SELECT * FROM employees WHERE salary < 60000;
```

Task 48

Display employees whose salary is between ₹45000 and ₹60000.

Commands:

```
SELECT * FROM employees WHERE salary BETWEEN 45000 AND 60000;
```

Task 49

Display employees from IT and salary greater than ₹50000.

Commands:

```
SELECT * FROM employees WHERE department='IT' AND salary > 50000;
```


Task 50

Display employees from HR or Finance.

Commands:

```
SELECT * FROM employees WHERE department='HR' OR department='Finance';
```

Task 51

Display employees not working in HR.

Commands:

```
SELECT * FROM employees WHERE department <> 'HR';
```

Task 52

Display employees from IT or Finance using IN.

Commands:

```
SELECT * FROM employees WHERE department IN ('IT','Finance');
```

Task 53

Display employees whose names start with 'S'.

Commands:

```
SELECT * FROM employees WHERE emp_name LIKE 'S%';
```

Task 54

Display employees whose names end with 'a'.

Commands:

```
SELECT * FROM employees WHERE emp_name LIKE '%a';
```

Task 55

Display employees whose names contain 'ar'.

Commands:

```
SELECT * FROM employees WHERE emp_name LIKE '%ar%';
```

Task 56

Display unique department names.

Commands:

```
SELECT DISTINCT department  
FROM employees;
```

Task 57

Display employees sorted by salary ascending.

Commands:

```
SELECT * FROM employees ORDER BY salary ASC;
```

Task 58

Display employees sorted by salary descending.

Commands:

```
SELECT * FROM employees ORDER BY salary DESC;
```

Task 59

Display employees sorted by employee name.

Commands:

```
SELECT * FROM employees ORDER BY emp_name;
```

Task 60

Display first 5 employee records using LIMIT.

Commands:

```
SELECT * FROM employees LIMIT 5;
```

Bonus DQL Tasks (Aggregate Functions)**Task 61**

Count total employees.

Commands:

```
SELECT COUNT(*) AS total_employees FROM employees;
```

Task 62

Find total salary paid to employees.

Commands:

```
SELECT SUM(salary) AS total_salary FROM employees;
```

Task 63

Find average salary.

Commands:

```
SELECT AVG(salary) AS average_salary FROM employees;
```

Task 64

Find highest salary.

Commands:

```
SELECT MAX(salary) AS highest_salary FROM employees;
```

Task 65

Find lowest salary.

Commands:

```
SELECT MIN(salary) AS lowest_salary FROM employees;
```

Task 66

Find number of employees in each department.

Commands:

```
SELECT department, COUNT(*) AS employee_count  
FROM employees GROUP BY department;
```

Task 67

Find total salary department-wise.

Commands:

```
SELECT department, SUM(salary) AS total_salary  
FROM employees GROUP BY department;
```

Task 68

Find average salary department-wise.

Commands:

```
SELECT department, AVG(salary) AS avg_salary  
FROM employees GROUP BY department;
```

Task 69

Display departments having more than 2 employees.

Commands:

```
SELECT department, COUNT(*) AS employee_count  
FROM employees GROUP BY department  
HAVING COUNT(*) > 2;
```

Task 70

Display department with highest average salary.

Commands:

```
SELECT department, AVG(salary) AS avg_salary  
FROM employees GROUP BY department  
ORDER BY avg_salary DESC LIMIT 1;
```